

VIVIEN PRAUD WALTER

R&D PROJECT MANAGER – AI, SOFTWARE & SCIENTIFIC INSTRUMENTATION

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PROFILE

Bilingual French-English R&D Project Manager with a PhD in Biophysics and nearly nine years of experience leading complex projects combining software development, AI, web platforms, XR and scientific instrumentation. End-to-end project ownership, from requirements to delivery: scoping, specifications, planning, funding, multidisciplinary team co-ordination, development, testing and validation. Equally comfortable in a leadership role or a hybrid position with hands-on technical contribution.

CORE SKILLS

Project management: scoping, requirements gathering, roadmaps and schedules, budgets and funding, risk, quality, reporting and client sign-off; multidisciplinary teams of up to 11 people; tailored Agile/Scrum, OpenProject and Affine.

Software & web: Python; TypeScript, JavaScript, React, Node.js, Express, MongoDB, Zod, API, Git/GitHub; LabVIEW; C; C++.

AI & data: TensorFlow, scikit-learn, 1D/3D CNNs, classification, image and signal processing, multi-modal generative AI and prompt engineering.

R&D and prototyping: laboratory setup, instrumentation, microfluidics, iSCAT, epifluorescence, UV-Vis, GROMACS, ImageJ, Fusion 360, 3D printing and Arduino/I2C.

PROFESSIONAL EXPERIENCE

Product Manager – LXP, AI & XR – i-magineXR

La Roche-sur-Yon, France | January 2024 – July 2026

- Led the development of a virtual campus: scoping, functional and technical specifications, schedule, trade-offs and sign-off with the client.
- Line and matrix management of 11 team members across web and XR development, UX/UI, research, communications and IT; weekly progress meetings, delegation through the Lead Developer and tailored Agile approach.
- Defined the project and wrote the full funding application, securing approximately €500k from OPCO Atlas.
- Designed and developed the TypeScript/Node.js/Express/MongoDB backend; TypeScript migration, Zod validation, authentication redesign, API keys, server-side pagination and resolution of blocking defects.
- Architected and developed a multi-model AI ecosystem integrating OpenAI, Claude, Gemini, Mistral and DeepSeek, including a processing queue, administration tools and project-side prompt engineering.

R&D Project Manager – Postdoctoral Research Associate – King's College London

Department of Engineering | London, United Kingdom | March 2021 – August 2023

- Built and commissioned a complex molecular communication laboratory: premises, equipment, procurement, safety, protocols and training for nine team members.
- Managed a team of five and €360k in EPSRC funding; contributed to one funding application.
- Designed an operational testbed validating chemical reaction-based signal processing, including filtering, saturation, amplification and encoding.
- Designed and built six open-source syringe pumps end to end using Fusion 360, 3D printing, Arduino, I2C, C++, and Python.
- Developed a 1D CNN in Python, TensorFlow and scikit-learn to estimate solution composition from UV-Vis signals; first author of Nature Communication and IEEE publications; contributed to discussions with Huawei and the preparation of patent application documentation.

R&D Project Manager – Postdoctoral Research Associate – King's College London

Department of Chemistry | London, United Kingdom | March 2018 – March 2021

- Led a €500k project funded by the BBSRC and co-funded by Oxford Nanopore Technologies; directly supervised three PhD researchers and two Master's students.
- Designed, assembled and qualified an iSCAT microscope, including component selection and LabVIEW development for control, acquisition and automation.
- Designed the experimental workflow: protocols, artificial membranes, gold nanoparticles and membrane proteins, experiments, and image and results analysis using Python, ImageJ and Java.
- Delivered a fully operational microscope; lead author of a Journal of Microscopy paper and co-author of publications in JACS and Nature Communications.

Independent Microscopy Consultant – Exclusive Contract – Vidya Health Ltd

London, United Kingdom | April 2020 – November 2020, temporary assignment during the COVID-19 pandemic

- Built an automated epifluorescence microscope for the analysis of medical samples, including autofocus design, technical specifications and LabVIEW software.
- Produced documentation and delivered training; the validated prototype was subsequently replicated on two additional microscopes.

R&D Project Manager – Postdoctoral Research Associate – University of Strasbourg

Strasbourg, France | October 2017 – March 2018

- Independently led the scientific and operational delivery of a project on lipid bilayer phase transitions, including planning, GROMACS simulations and Bash/SLURM cluster computing.
- Developed machine-learning approaches followed by a TensorFlow 3D CNN; lead developer of the open-source Python module MLLPA and author of its website and documentation.
- First author and lead writer of three publications in PCCP, Journal of Computational Chemistry and BBA Biomembranes.

Independent Python, Data Analysis & Generative AI Consultant

Ad hoc assignments since 2023 – activity currently paused

- Completed four assignments covering consulting, prompt-engineering training, classification and data analysis, and Python development or integration using scikit-learn, TensorFlow and PyTorch.

HIGHER EDUCATION TEACHING

King's College London – Lecturer, 2023: designed and delivered the Statistical Thermodynamics course in English to nearly 100 third-year undergraduate.

ECPM – Visiting Lecturer, 2021-2023: designed a complete Python, machine-learning and deep-learning module applied to chemistry for approximatively 90 Master's-level engineering students.

EDUCATION

PhD in Biophysics: University of Strasbourg, 2014-2017 – Lipid membrane interactions with self-assembling cell-penetrating peptides.

MSc Physics – Materials and Nanoscience: University of Strasbourg, 2013-2014.

MSc Physics and Chemistry – Education Science: University of Strasbourg, 2011-2013.

SELECTED PUBLICATIONS

- Real-time signal processing via chemical reactions for a microfluidic molecular communication system – Nature Communications, 2023; first author and lead writer.
- CNN-based detection of mixed-molecule concentrations in Molecular Communication – IEEE T-MBMC, 2026; first author and lead writer.
- Spatial light modulation for interferometric scattering microscopy – Journal of Microscopy, 2024; first author and lead writer.
- Ripple-like instability in the simulated gel phase of finite size phosphocholine bilayers – BBA Biomembranes, 2021; first author and lead writer.

ADDITIONAL INFORMATION

Languages: native French; bilingual professional proficiency in English, with five years in London and university-level teaching delivered in English.

Mobility: Full driving European license and own vehicle, available for on-site, hybrid or remote work; immediately available, permanent position preferred